

SSPA ultimate cross-topic killer questions

T1-T10 Full Trap Mix · Most difficult lectures throughout the school year · 65 minutes · 1-to-3 Online Lesson

Positioning: The most difficult **handout for this school year** |||SEP|||. Simulate the real SSPA sub-test finale question style - each question contains 3+ trap types

□ T1-T10 full series - each question contains traps, training students' "trap detection

radar" SSPA correlation:

● Extremely high frequency

The last 5-8 questions in the actual sub-test are all cross-topic comprehensive questions, which are highly consistent with the question types of this handout

Prerequisite knowledge:

All topics in the whole school year (multiple

digits · decimals · fractions · area · volume · equations · data · word questions)

Class objectives:

① Cultivate trap detection ability ② Parallel processing of multiple traps ③ Maintain calculation accuracy under high pressure

Difficulty setting:

No basic questions in the whole volume - all 🟢 Advanced or 🟡 Challenge level. This is preparation for SSPA actual combat.

There are no basic questions in the whole volume - all 🟢 advanced or 🟡 challenge levels. This is preparation for SSPA actual combat.

Student Name:

Class:

Date:

Time Spent:

I. Warm-up start - trap radar test (total 3 questions, 5 minutes)

#	Question	Difficulty	Working Space
1	Calculation: $\frac{2}{3} \times 0.75 + 1.25 = ?$ (First find out how many traps are hidden in this question! Hint: decimal point + fraction + order of operations)		
2	A triangle has a base of 0.25 m and a height of 40 cm. What is the area in square centimeters? (Unit trap + formula trap)		
3	A rectangular water tank: $50\text{cm} \times 30\text{cm} \times 40\text{cm}$. Contains $\frac{3}{4}$ water. What is the volume of water in liters? (Three hits of volume + fraction + unit conversion)		

 **Our core philosophy: "When you see a question, don't worry about it. Scan where the traps are. Mark the traps before writing and defeat them one by one."**

II. Core Knowledge

Knowledge point 1: Calculation killer questions  - Each question contains at least two T1 (advance and retreat) + T2 (decimal point) + T9 (fraction)

- ① **Features** |||SEP|||: The questions involve multi-digit operations on integers, decimal point positioning, fraction generalization/reduction Strategy |||SEP|||: First unify the format (fraction first), then calculate in sections, and check each step
- ② **SSPA Live** |||SEP|||: The last three questions of paper 1 calculation questions are usually such mixed questions, and the score loss rate in Hong Kong > 60% The deadliest combination |||SEP|||: large number advance and retreat (T1) + decimal division shift (T2) + fractional reduction (T9) = three consecutive traps
- ③ **SSPA Live**: The last three calculation questions in Paper 1 are usually mixed questions of this type, and the score loss rate in Hong Kong is > 60%
- ④ **The deadliest combination**: Large number advance and retreat (T1) + decimal division shift (T2) + fractional reduction (T9) = triple trap

❑ **Killer example (T1+T2+T9 three consecutive hits)**

Calculation: $(125.5 + 874.5) \times \frac{2}{3} \div 0.5 = ?$

 **Common ways to die**

 **Correct solution**

125.5+874.5=990 (carry leakage),
 990 × 2/3 = 1980/3 = 660 (regarded
 as × 3), 660 ÷ 0.5 = 330

Two traps in a row: wrong carry + fraction
 multiplication, one wrong step and the next wrong
 step

$$1000 \times \frac{2}{3} \div 0.5 = \frac{2000}{3} \times 2 = \frac{4000}{3} = 1333\frac{1}{3}$$

Addition check: 125.5+874.5=1000 ✓; ÷ 0.5 = × 2;
 Fraction multiplication: 1000 × 2/3 × 2 = 4000/3

Knowledge Point - Killer Questionspractice (trap included in perquestion annotation,
 total 12 questions - all 🌳/🏠 level)

#	Question	Difficulty	Working Space
K1-1	T1+T9 $(45678 + 12345) \times \frac{1}{3} = ?$ (Large number addition after carry × fraction)	🌳	
K1-2	T2+T9 $3.14 \times \frac{5}{6} + 0.86 = ?$ (Decimal × fraction, decimal places of product + fraction reduction)	🌳	
K1-3	T1+T2 $128.5 \times 64 - 256.8 \div 0.4 = ?$ (Carry for large number multiplication + decimal division)	🌳	
K1-4	T2+T9+T1 $(0.125 \times 8000) + (\frac{5}{8} \times 0.64) = ?$ (Three traps are launched in unison)	🏠	

#	Question	Difficulty	Working Space
K1-5	T1+T9 $(9999 + 5678) \times \frac{3}{7} = ?$ (Carry to tens of thousands + fractional result reduction)	🌳	
K1-6	T2+T9 $0.075 \div \frac{3}{8} + 0.04 = ?$ (Decimal ÷ Fraction → × Reciprocal → Number of decimal places)	🏠	
K1-7	T1+T2 $725.6 \times 1.5 - 488.4 = ?$ (Carry for multiplication + number of decimal places + carry for subtraction)	🌳	
K1-8	T9+T2+T1 $\frac{2}{3} \times (456 \div 0.25) = ?$ (Carry for decimal division within parentheses + multiplication of outer fractions)	🏠	

K1-9	T2+T9 $0.375 + \frac{5}{8} - 0.125 \times 4 = ?$ (First multiply and divide, then add and subtract + mixed fractions and decimals)		
K1-10	T1+T9+T2 $(15000 - 8765) \times \frac{2}{5} \div 0.2 = ?$ (Five-step calculation: subtraction abdication $\rightarrow$ $\times$ fraction $\rightarrow$ $\div$ decimal $\rightarrow$ all traps)		
K1-11	T2+T9 $1.25 \times (\frac{3}{4} + \frac{1}{6}) \div 0.25 = ?$ (Common fraction of fractions in brackets $\rightarrow$ $\times$ decimal $\rightarrow$ $\div$ decimal)		
K1-12	T1+T2+T9 $(888 \times 999) \times 0.125 + \frac{1}{8} = ?$ (SSPA level ultimate difficulty: multiplication of very large numbers + decimals + fractions)		

Knowledge point 2: Application killer questions - Each question contains T3 (word comprehension) + T7 (unit conversion) + at least one other trap

- ① **Features** |||SEP|||: Long text questions + multiple units + interfering information + multi-step reasoning
Strategy |||SEP|||: Circle keywords $\rightarrow$ draw line graphs $\rightarrow$ mark units $\rightarrow$ column formulas $\rightarrow$ step by step calculation $\rightarrow$ check units
- ② **40% Fatal combination** |||SEP|||: Understanding reversal (T3) + Unit disunity (T7) + Careless calculation (T1/T2/T9)
- ③ **SSPA Live**: The last 2 questions of paper 2, usually 6-8 points, the average score rate in Hong Kong is < 40%
- ④ **deadly combination**: Understanding reversal (T3) + Unit inconsistency (T7) + Careless calculation (T1/T2/T9)

Killer example (T3+T7+T9)

A rope is $\frac{5}{2}$ meters long. I used $\frac{1}{4}$ on the first day, the remaining $\frac{2}{3}$ on the second day, and 75 cm on the third day. How many centimeters are left at the end?

Knowledge Point 2 Killer Questions practice (total 12 questions - all / levels, including complete answer sentences need to find)

#	Question	Difficulty	Working Space (column $\rightarrow$ calculate $\rightarrow$ answer sentence, both are indispensable)
K2-1	T3+T7 A rectangular field: 0.15 km long and 80 m wide. How many hectares is the area? (1 hectare =		

	10000m ² , 1km = 1000m, first convert everything into meters)		
K2-2	T3+T9+T7 One bottle of juice 2.5 liters. I drank $\frac{1}{5}$ for breakfast, 350 ml for lunch, and the rest of $\frac{1}{2}$ for dinner. How many milliliters are left at the end?		
K2-3	T3+T2+T7 A water tank can hold 12.5 liters of water. If 8750 ml is injected now, how many liters can be injected? (Note that "okay" = capacity - existing, the units must be unified)		
K2-4	T3+T7+T9 Xiao Ming has \$360. I used $\frac{1}{3}$ to buy books, then used the remaining 0.25 to buy stationery, and finally used \$45.50 to buy snacks. How many dollars are left?		

#	Question	Difficulty	Working Space
K2-5	T3+T7+T1 A packet of sugar weighs $\frac{3}{4}$ kg. After using 280g, divide the remainder into 95g bags. How many bags can be divided into at most? How many grams are left? (Multiple steps + unit conversion + packaging issues)		
K2-6	T3+T9+T2 The engineering team completed the project in three days. Complete $\frac{2}{5}$ on the first day, 0.05 more on the second day, and the rest on the third day. What fraction of the project was completed on the third day?		
K2-7	T3+T2+T7 Original price for a coat is \$680. First increase the price by 15% (i.e. $\times 1.15$), and then discount it by 20%. Is the final selling price more expensive or cheaper than the original price? How much difference?		
K2-8	T3+T7+T9+T2 A rectangular swimming pool: 25 meters long, 12 meters wide and 1.5 meters deep. Contains $\frac{3}{5}$ water. How many liters of water are there in the pool? (1m ³ =1000L, four traps in a row)		
K2-9	T3+T9+T2 A book has 480 pages. I read $\frac{1}{4}$ on the first day, the remaining 0.6 on the second day, and 85 pages on the third day. How many pages are left to read?		

K2-10	T3+T7+T1 The rectangular garden is 0.08 km long and 45 m wide. The gardener walked 5 times around the garden, how many kilometers did he cover in total? (Perimeter $\times$ number of circles, please note the unit: km $\leftrightarrow$ m)		
K2-11	T3+T7+T9+T2 Inside of a water tank: 60cm $\times$ 40cm $\times$ 50cm. The original water height was 30cm. An irregular stone (volume 7200cm ³) was placed and completely sunk. How high will the new water level be? Answer with rice.		
K2-12	T3+T2+T9+T7 Xiao Ming saves \$500. The first interest class used SEP , the second used the remaining 0.375, and the third used \$85. How much is left? If you use the remaining $\frac{2}{5}$ to buy books, how much did you spend on the books? (Four traps + double layer remaining) $\frac{3}{4}$ How much did it cost to buy the book? (Four traps + double layer remaining)		

Knowledge point 3: Geometry killer questions - Each question contains T4 (confusion of geometric formulas) + T5 (solid geometry) + at least one other trap

- ① **Features |||SEP|||: Composite graphics + three-dimensional + unit trap + calculation trap** Strategy |||SEP|||: Label all known quantities first $\rightarrow$ write the required formulas $\rightarrow$ unify the units $\rightarrow$ algebraic calculation $\rightarrow$ check the rationality of the units
- ② **SSPA Live |||SEP|||: The highest score loss rate for geometry questions in the sub-test, because there are many formulas and hidden unit traps** Fatal combination |||SEP|||: Triangle $\div 2$ forget (T4) + volume representative surface area formula (T5) + cm/m mixed (T7)
- ③ **SSPA Live:** The geometry questions in the sub-test have the highest score loss rate because there are many formulas and hidden unit traps
- ④ **deadly combination:** Triangle $\div 2$ Forget (T4) + Volume representative surface area formula (T5) + cm/m Mix (T7)

Killer Example (T4+T5+T7)

The interior of a rectangular water tank: 0.5 m long, 30 cm wide, and 40 cm high. After filling the water the water level reaches 25 cm. Another triangular prism with a base area of 200cm² is placed (completely sunk) and the water level rises to 32cm. What is the height of the triangular prism in centimeters?

Knowledge Point 3 Killer Questionspractice (total 12 questions - all / levels)

#	Question	Difficulty	Working Space
---	----------	------------	---------------

K3-1	T4+T7 A triangle has a base of 0.4 meters and a height of 25 centimeters. What is the area in square centimeters? (Bottom $\times$ height $\div$ 2, note: 0.4m=40cm)		
K3-2	T4+T5+T7 A trapezoidal garden bed: upper base 8 meters, lower base 12 meters, height 5 meters. A layer of soil 15 cm thick should be spread on top. How many cubic meters of soil are needed? (Trapezoidal area $\times$ thickness = volume, thickness is converted to meters first)		
K3-3	T4+T5+T2 A cube has a side length of 0.5 meters. What is the surface area in square meters? How many cubic meters is the volume? How many liters is the volume?		
K3-4	T4+T1+T7 Composite graphic: a rectangle (12m $\times$ 8m) with a triangle (base 12m and height 5m) stacked on top. Total area=? (Rectangle + triangle, please note that the units are consistent)		

#	Question	Difficulty	Working Space
K3-5	T5+T7+T9 A rectangular water tank: 50cm $\times$ 40cm $\times$ 60cm. Original Water $\frac{1}{3}$ Box. Put in a rectangular iron block (20cm $\times$ 15cm $\times$ 10cm) and sink it completely. How many centimeters will the water level rise?		
K3-6	T4+T5+T7+T2 A parallelogram has a base 0.25 m and a height 18 cm. A cuboid is formed with its base and height (base $\times$ height is the base, thickness is 5 cm). Calculation: area of parallelogram + volume of cuboid. (Four traps!)		
K3-7	T4+T5+T9 A cuboid: length 24 cm, width equal to the length SEP , height equal to the width SEP . Volume=? Surface area=? (First find the width and height, fraction operation + geometric double kill) $\frac{1}{3}$, high is wide $\frac{1}{2}$. Volume=? Surface area=? (First find the width and height, fraction operation + geometric double kill)		
K3-8	T4+T7+T1 A trapezoid: the upper base is 3.5 meters, the lower base is 6.5 meters, and the height is 400 centimeters. How many square meters is the area? Use a fence to surround the four sides of the trapezoid. What is the minimum length of the fence? (Find the area first and then find the perimeter, but the perimeter of the trapezoid needs to know two waists - assuming it is an isosceles trapezoid, the waist length is calculated using Pythagorean theorem: half-base difference = 1.5m, waist = $\sqrt{(1.5^2 + 4^2)} = \sqrt{18.25} \approx 4.27$ m)		

K3-9	T5+T2+T7 A fish tank: internal dimensions $0.8\text{m} \times 0.5\text{m} \times 0.6\text{m}$. Once filled, scoop out the water using a 2.5 liter bucket. How many buckets can be scooped at most? (Volume of fish tank $\div$ capacity of each barrel, note $\text{m}^3 \rightarrow \text{L}$)		
K3-10	T5+T4+T7+T9 L-shaped composite three-dimensional: bottom layer $10\text{cm} \times 8\text{cm} \times 4\text{cm}$, upper layer $4\text{cm} \times 8\text{cm} \times 3\text{cm}$ (placed at one end of the bottom layer). Find: total volume, total surface area (note: joint surfaces are not counted as surface area). Four traps!		
K3-11	T4+T5+T7+T2 A cube has a side length of 0.25 meters. Saw it all into small cubes with 5 cm sides. How many pieces can it be sawed into? How much more is the total surface area of all the small cubes than the original large cube?		
K3-12	T4+T5+T2+T7+T9 The length of the base of a triangle = the length of the side of a cube (0.4 meters). Height of triangle = volume of cube $\div$ area of base of cube. Area of triangle = ? cm^2 . Then use the area of this triangle as the base area and add the height of 0.15 meters to form a triangular prism volume = ? cm^3 . (Five traps ultimate geometry question)		

Knowledge point 4: The ultimate mixed killer question - no classification, purely comprehensive, each question contains 4+ traps

- ① **Features** |||SEP|||: No obvious topic tags - calculations + text comprehension + units + geometry are all mixed together Strategy |||SEP|||: Mark all traps first $\rightarrow$ decompose into sub-questions $\rightarrow$ solve one by one $\rightarrow$ merge answers
- ② **SSPA Live** |||SEP|||: This is the closest to the real SSPA finale question (the last 2-3 of each volume of Volume 1 and Volume 2) **Difficulty of questions** Completion standard |||SEP|||: Can complete at least 5/9 questions correctly within the time limit, that is, you have the ability to achieve full SSPA score
- ③ **SSPA Live**: This is the difficulty closest to the real SSPA final questions (the last 2-3 questions of Paper 1 and Paper 2)
- ④ **Completion standards**: Able to complete at least 5/9 questions correctly within the time limit, that is, you have the ability to achieve SSPA full score

 **WARNING: The following 9 questions are among the most difficult this school year. Each question contains 4 or more traps. It is recommended to list all the traps on a scratch paper before starting the calculation.**

Ultimate mixed practice (total 9 questions - full  level, per question ≥ 4 trap)

#	Question	Difficulty	Working Space (first column trap $\rightarrow$ again calculate)
---	----------	------------	---

End 1	T2+T9+T3+T1 A farmer has 250.5 kg of rice. He sells $\frac{3}{5}$ and gives the remaining 0.25 to his neighbor. Finally he divided the remaining rice into 12.5kg bags. Q: How many bags can be divided into at most? How many kilograms are left? (Four traps: fractions + decimals + remaining understanding + division by pack)		
End 2	T4+T5+T7+T2 A rectangular swimming pool: 0.05 km long, 25 m wide, average water depth 1.8 m. There is water in the pool $\frac{4}{5}$ Full. How many minutes will it take to drain the pool using a pump that pumps 450 liters of water per minute? (Pool volume → water volume → time, all traps)		
End 3	T3+T9+T2+T7+T1 Xiao Ming, Xiao Hua, and Xiao Mei share a lump sum of \$1,500. Xiao Ming gets SEP , Xiao Hua gets 0.75 times what Xiao Ming gets, and Xiao Mei gets the rest. Xiaomei saves $\frac{2}{5}$ of her money and donates the rest to charity. Question: How much did she donate? (Five traps, super long question) $\frac{1}{3}$ Save it and donate the rest to charity. Question: How much did she donate? (Five traps, super long question)		

#	Question	Difficulty	Working Space
End 4	T5+T4+T7+T9+T2 An L-shaped composite solid: the bottom cuboid A (60cm × 40cm × 25cm), and the upper cuboid B (20cm × 40cm × 15cm) are placed above the right end of A. Find: (a) Total volume =? cm ³ (b) Total surface area =? cm ² (joint surfaces are not counted) (c) If the entire solid body is sunk into a water tank (the bottom area of the tank is 5000cm ²), how many centimeters will the water level rise? (Five Trap Geometry Ultimate Question)		
End 5	T2+T9+T3+T7+T1 The original water capacity of a water tank is 45.5 liters. First withdraw SEP , then inject 8750 ml, then withdraw the remaining 0.4, and finally inject 12.25 liters. How many milliliters of water is in the tank now? (Five traps: fractions + decimals + mixing units + remaining understanding + large number operations) $\frac{1}{5}$, inject another 8750 ml, withdraw the remaining 0.4, and finally inject 12.25 liters. How many milliliters of water is in the tank now? (Five traps: fractions + decimals + mixing units + remaining understanding + large number operations)		

End 6	T4+T5+T7+T3+T9 A trapezoidal garden: upper base 8.5 meters, lower base 11.5 meters, height 6 meters. $\frac{3}{5}$ Plant roses in the flower garden. How many square meters is the area where roses are planted? If 4 roses are planted per square meter, how many roses are planted in total? If each rose requires 250 ml of water and is watered once a day, how many liters of water will be required in a week? (Five trap long chain questions - area → fraction → multiplication → unit conversion → time)		
End 7	T3+T2+T9+T1+T7 Project: Team A will complete the project in 20 days, and Team B will complete the project in 30 days. After the two teams work together for 5 days, Team A takes a break and Team B completes the rest alone. How many more days will it take for Team B? (Engineering problem - first find their respective efficiencies: $A = \frac{1}{20}/\text{day}$, $B = \frac{1}{30}/\text{day}$. Five traps)		
End 8	T4+T5+T2+T7+T9 A rectangular glass water tank: outer dimensions $52\text{cm} \times 32\text{cm} \times 42\text{cm}$, glass thickness 1cm (without cover). Find: (a) Internal capacity =? Liters (round up) (b) How many cm^2 of glass are needed to make this tank? (c) If a stone with a volume of 4800cm^3 is placed in $\frac{3}{4}$ water, will the water overflow? If it overflows, how much will it overflow? (Six trap practical questions - internal and external dimensions + volume + area + fraction + overflow judgment)		
end 9	T1+T2+T3+T7+T9+T4 The ultimate challenge: a circular garden 14 meters in diameter ($\pi \approx \frac{22}{7}$). There is a square pool in the center of the garden (3.5 meters on a side). $\frac{3}{4}$ of the garden is paved with grass, and the rest is paved with stone bricks. Grass is \$45.50 per square meter and stone tiles are \$68.25 per square meter. The cost of paving grass + the cost of paving stone bricks =? How does the total cost compare to \$25,000? (SSPA level ultimate question - area of circle + area of square + fraction + multiplication of decimals + comparison of large numbers + six traps)		

V.SSPA Killer Questions Summary of Survival Rules

①

Detect traps

After reading the question, mark all possible pitfalls

②

unified format

Unify all fractions and decimals

③

Segment calculation

No skipping steps in each step of verification

④

check answer

Unit and rationality backward verification

 **SSPA killer mantra: "Trap detection is the first step, uniform format is the basic skill, segment calculations are correct, and check the answers to ensure full marks."**

VI. The Lesson core common errors summary

Self-examination in this hall (tick after completion)

☐ I know the pitfalls of solving each knowledge point

☐ I can complete  basic questions independently

☐ I can challenge  advanced questions

☐ I remember the formula

Review of Learning Objectives - After completing this lesson you should be able to:

☐ Identify all trap types in our hall

☐ Solve  basic questions independently (100% correct)

☐ Challenge  Advanced questions (80%+ correct)

☐ Explain the lesson formula to classmates

#	common error	Correct Approach
1	A certain trap was missed in the multi-trap question, resulting in local errors.	For each question, pause for 10 seconds to scan and mark all possible traps before starting the calculation.
2	In the word problem, the direction of "remainder" is reversed SEP , using the original number instead of the remainder , use the original number instead of the remainder	Draw a line segment diagram: update the remaining amount after each "remainder" and use it as the base for the next step
3	The units are not uniformly substituted into the formula SEP : cm and m are used together, L and mL are used together. : cm and m are used together, L and mL are used together	All lengths are first unified into the same units; 1m=100cm, 1L=1000mL, 1m ³ =1000L
4	The area of a triangle is $\div 2$ / the trapezoid formula is misremembered	Write formula $\rightarrow$ Algebra $\rightarrow$ Calculate; triangle $A=bh/2$, trapezoid $A=(a+b)h/2$
5	Composite three-dimensional joint surface repeated calculation of surface area	The joint surface does not belong to the external surface and is deducted from the total
6	Efficiency is reversed in engineering problems SEP : The more days, the lower the efficiency: The longer the number of days, the lower the efficiency.	Efficiency = $1 \div$ number of days; cooperation efficiency = the addition of efficiencies; time = workload $\div$ efficiency
7	The answer misses a unit, a sentence, or the unit is written incorrectly.	SSPA scoring: 1-2 step points will be deducted for missed sentences; 0 points for wrong units